Experiment No:02

Title: Perform recursive Function binary search using C.

Aim:
To write a C program to perform binary search using recursion on a sorted array of integers.

Mapped to Course Objective:
To employ the different data structures to find the solutions for specific problems.

Theory:

Binary Search is an efficient algorithm for finding an element in a sorted array by repeatedly dividing the search interval in half.

Recursive Binary Search breaks the problem into smaller subproblems. Each recursive call eliminates half of the array until the element is found or the subarray size becomes zero.

Important Note: Binary search only works on sorted arrays . **Pointers and strings:** Strings can be stored at some memory location and assign the address of the string in a char pointer. This is shown as below:

```
char str[]=”Hello”;
char *p=”Hello”;
```

We can assign a char pointer to another char pointer.

Algorithm:

1. **Input: A sorted array arr[], the element to search key, low index low, and high index high.**
2. **Check base case: If low > high, return -1 (element not found).**
3. **Compute mid index: mid = (low + high) / 2**
4. **If arr[mid] == key → Element found, return mid**
5. **If key < arr[mid] → Search in the left half: call recursively with high = mid - 1**
6. **Else → Search in the right half: call recursively with low = mid + 1**

```c
#include<stdio.h>

int binary(int arr[],int low,int high,int key){
    if(low>high){
        return -1;
    }

    int mid=(low+high)/2;

    if(arr[mid]==key){
        return mid;
    }
    else if(key<arr[mid]){
        return binary(arr,low,mid-1,key);
    }
    else{
        return binary(arr,mid+1,high,key);
    }
}

int main(){
    int arr[100],n,key,i,result;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter sorted array elements: ");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }

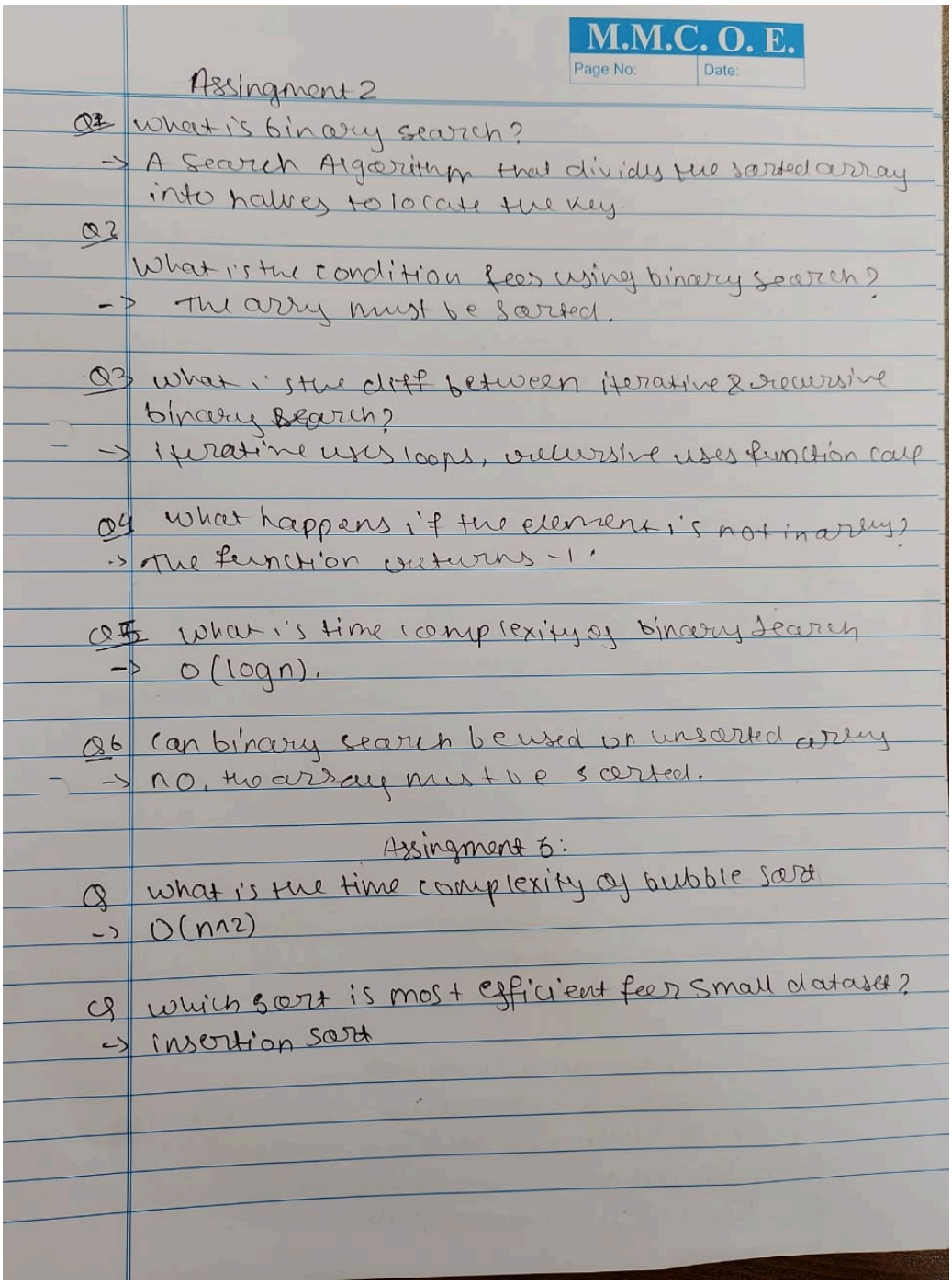
    printf("Enter element to search: ");
    scanf("%d",&key);

    result=binary(arr,0,n-1,key);

    if(result==-1){
        printf("Element not found");
    }
    else{
        printf("Element found at position %d",result+1);
    }

    return 0;
}
```

Output:
Enter number of elements: 5
Enter sorted array elements: 10 20 30 40 50
Enter element to search: 30
Element found at position 3



M.M.C. O. E.
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Assignment 2

Q1 what is binary search?
-> A search Algorithm that divids the sorted array into halves to locate the key.

Q2 What is the condition for using binary search?
-> The array must be sorted.

Q3 what is the diff between iterative & recursive binary search?
-> iterative uses loops, recursive uses function call

Q4 what happens if the element is not in array?
-> The function returns -1.

Q5 what is time complexity of binary search
-> O(logn).

Q6 can binary search be used on unsorted array
-> no, the array must be sorted.

Assignment 5:

Q what is the time complexity of bubble sort
-> O(n^2)

Q which sort is most efficient for small dataset?
-> insertion sort